

14-3e Why the Long-Run Supply Curve Might Slope Upward

So far, we have seen that entry and exit can cause the long-run market supply curve to be perfectly elastic. The essence of our analysis is that there are a large number of potential entrants, each of which faces the same costs. As a result, the long-run market supply curve is horizontal at the minimum of average total cost. When the demand for the good increases, the long-run result is an increase in the number of firms and in the total quantity supplied, without any change in the price.

There are, however, two reasons that the long-run market supply curve might slope upward. The first is that some resources used in production may be available only in limited quantities. For example, consider the market for farm products. Anyone can choose to buy land and start a farm, but the quantity of land is limited. As more people become farmers, the price of farmland is bid up, which raises the costs of all farmers in the market. Thus, an increase in demand for farm products cannot induce an increase in quantity supplied without also inducing a rise in farmers' costs, which in turn means a rise in price. The result is a long-run market supply curve that slopes upward, even with free entry into farming.

A second reason for an upward-sloping supply curve is that firms may have different costs. For example, consider the market for painters. Anyone can enter the market for painting services, but not everyone has the same costs. Costs vary in part because some people work faster than others and in part because some people have better alternative uses of their time than others. For any given price, those with lower costs are more likely to enter than those with higher costs. To increase the quantity of painting services supplied, additional entrants must be encouraged to enter the market. Because these new entrants have higher costs, the price must rise to make entry profitable for them. Thus, the long-run market supply curve for painting services slopes upward even with free entry into the market.

Notice that if firms have different costs, some firms earn profit even in the long run. In this case, the price in the market reflects the average total cost of the *marginal firm*—the firm that would exit the market if the price were any lower. This firm earns zero profit, but firms with lower costs earn positive profit. Entry does not eliminate this profit because would-be entrants have higher costs than firms already in the market. Higher-cost firms will enter only if the price rises, making the market profitable for them.

Thus, for these two reasons, a higher price may be necessary to induce a larger quantity supplied, in which case the long-run supply curve is upward-sloping rather than horizontal. Nonetheless, the basic lesson about entry and exit remains true. *Because firms can enter and exit more easily in the long run than in the short run, the long-run supply curve is typically more elastic than the short-run supply curve.*

QuickQuiz

6. In the long-run equilibrium of a competitive market with identical firms, what are the relationships among price P , marginal cost MC , and average total cost ATC ?
- a. $P > MC$ and $P > ATC$.
 - b. $P > MC$ and $P = ATC$.
 - c. $P = MC$ and $P > ATC$.
 - d. $P = MC$ and $P = ATC$.
7. In the short-run equilibrium of a competitive market with identical firms, if new firms are getting ready to enter, what are the relationships among price P , marginal cost MC , and average total cost ATC ?
- a. $P > MC$ and $P > ATC$.
 - b. $P > MC$ and $P = ATC$.
 - c. $P = MC$ and $P > ATC$.
 - d. $P = MC$ and $P = ATC$.
8. Suppose pretzel stands in New York City are a perfectly competitive market in long-run equilibrium. One day, the city starts imposing a \$100 per month tax on each stand. How does this policy affect the number of pretzels consumed in the short run and the long run?
- a. down in the short run, no change in the long run
 - b. up in the short run, no change in the long run
 - c. no change in the short run, down in the long run
 - d. no change in the short run, up in the long run

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